BACKGROUND AND AIMS – *Introduction*

Cardiovascular Diseases (CVD), and more particularly, those associated with vessel stenosis, occlusion, and atherosclerosis, continue to prevail as the leading causes of global death. Implantable solutions (such as artificial vascular conduits) are generally not sufficiently compatible for the conditions of the body, interact counter-effectively with bodily processes, and subsequently are not suitable as long-term treatment methods. In consideration of trends in health conditions such as CVDs, and the large number of vessel-related medical procedures such as bypass surgery, angioplasties, and endarterectomy, the capacity to develop strong, viable, long-term vessel substitutes is of substantial value. On top of existing medical demands for artificial vasculature, the ability to develop such conduits expands potential for other biotechnologies such as bioprinting pre-vascularized artificial tissues, organs and in preclinical drug assessments.

CVDs remain the leading cause of death; the most common form of which is ischemic heart disease, or Coronary Heart Disease (CHD), which claims over 370,000 lives annually; with general heart disease accounting for approximately 610,000 deaths annually. CHD affects around 500-600 million people today and is continually increasing with time (Roth et al., 2020). CHD, most often characterised by atherosclerosis, entails the build-up of fatty plaque inside blood vessels which impede the flow of blood and perfusion of tissues resulting in hypoxia and tissue degeneration. Current surgical treatments for atherosclerosis and CHD include bypass surgery, where a vessel, often a periphery artery, or (if a healthy donor vessel cannot be identified) an artificial conduit, is substituted in place of the vessel of concern and is sutured to the native vessel system. This replaces the previously occluded vessel, allowing blood to flow freely, relieving blood pressure, and restoring blood supply.

CURRENT TREATMENTS - Drawbacks of current treatments and gaps in research

Naturally Sourced Vessels (NSVs) and Artificial Blood Vessels (ABVs) characterise essentially all surgical treatment options for vascular occlusion conditions. Both are at different developmental stages, and both constitute their own sets of advantages and disadvantages. Non-surgical treatment strategies exist beyond the scope of this report. Established treatment methods face challenges such as the safe sourcing vessels, the longevity of treatment, minimisation of health risks (commonly having to do with vessel occlusion) and the inability to develop vessel treatment for the long term.

Autografts (NSV) - Most coronary bypass vessel surgeries utilise grafts which are autologously sourced from (one of) the saphenous vein in the leg, an internal thoracic artery, or from the radial artery. A common, and often-deemed optimal donor vessel, particularly for a coronary artery bypass graft, is the internal thoracic artery, as per its viability and sustainability in long term (Martínez-González et al., 2017). Morphological similarities, such as comparable elasticity, and high permeability of this graft are noted as key factors which decrease the probability of atherosclerosis and hyperplasia (growth of smooth muscle cells on the inner-intima, impairing flow through the lumen of the vessel). The build-up of plaque in atherosclerotic vessels is generally not localized to a singular region but is distributed variably throughout the vascular system (Pan et al., 2022), due to the nature of the pathology. Subsequently, one of the prevailing challenges associated with autologously sourced vessels is that the selection of donor vessels is limited to the vessels which remain healthy, comparably structured, and relevant for vascular application (Hoenig et al., 2005). Other autograft-generalised risks such as the potential for donor site complications and the variability in the condition and longevity of grafts across patients, remain relevant. Donor selection

is also constraint by the fact that venous grafts, as opposed to arteries, are more prone to thrombosis, occlusion, and aneurysm (Verma et al., 2004), thus less used for surgeries.

Artificial Conduits (ABV) - Readily available synthetic grafts, in clinical applications, have shown moderate success at the macrovascular scale, however, reliable fail at smaller diameters (<6 mm inner diameter) (Lovett et al., 2007). Reasons for conduit failure most often include thrombosis, calcification, intimal hyperplasia, infections, poor biomechanical/biochemical/biological mimicry, and poor longevity in vivo (Sologashvili et al., 2019). Most clinically used ABVs are made from one, or a combination of polymers: expanded polytetrafluoroethylene (ePTFE) also known as Gore-Tex, or polyethylene terephthalate (PET) also known as Dacron, most often. Others include polyurethane (PU) and Polycaprolactone (PCL). These have proven as semi-viable options for short term applications yet fail commonly due to thrombosis and neointimal hyperplasia (Filipe et al., 2018). Commercial-grade ABVs are made from made from these polymers which happen to be strong, flexible in manufacturing, and easy to store, however chemically inert, and often stiff. Such properties do not mimic those of native tissue and are unideal for integration with the native vasculature, leading many artificial vessel models to fail, especially at smaller diameters (<6mm) (Moore et al., 2022). These alternatives fail to replicate important characteristics of the native tissue and thus react poorly with and amongst biological mechanisms. Generally, hydrophobicity and rough internal surfaces of conduits interacts adversely with endothelial cells, leading to the development of blood clots and incomplete cellular integration (Bao et al., 2020; Jaffer et al., 2015). Though many efforts have tried to resolve such insufficiencies, harsh compromises are often made in time and cost efficiency, especially in new manufacturing and culturing methods.

Solution Proposal - A two-layer electro spun conduit. The inner layer is made of coaxially electro spun PU and PCL, the respective properties of which counter each other to controllably slow cell proliferation, inhibiting hyperplasia and thrombosis. The outer layer is made of a more loosely-electro spun Ultra-High Molecular Weight Polypropylene (UHMWPE), a strong, hydrophobic, biocompatible polymer which prolongs the durability of the scaffold, aided in this by its lubricative properties. “Loosely electro spun” refers to the fact that this layer is (by changing collecting drum rotational speed) spun less tightly than the inner, facilitating a degree of circumferential dilation capability in the conduit during changes in blood pressure/ volume, to decelerate wear. UHMNWPE is widely employed in medicine, FDA-approved as an implantable, and is characterised by its very high-wear resistance, ductility, and biocompatibility, (Hussain et al., 2020) and durability.

LITERATURE REVIEW - *Research Developments, Strategies, and Market Products*

Artificial conduits can be categorised as either Naturally Sourced or an Artificial/ Tissue-Engineered. The former deems that the source of the vascular substitute is the patient’s own tissues (autograft), another human subject (allograft), or an animal (xenograft). In the instance that a naturally-source graft is treated through processes, they will remain considered as a naturally sourced vessel in this report. Artificial conduits entail alternatives synthesised through application of the range of manufacturing, cellular, and treatment processes available to the scientific space.

[Other] Naturally Sourced Vessels – *Allografts*

While not as common a source as autologously sourced vessels, allografts have been trailed and tested in clinical context. Mechanisms such as cryogenic freezing and cellular treatment processes have been investigated with the objective of improving the viability of allogenic vessels and

minimising immune rejection complications. In 2017, Harlander-Locke et al had demonstrated, that following 51±24 months post-operation, a 37% thrombosis (early and late) rate in a sample of 457 patients undergoing haemodialysis. The allograft sources for these patients were cryopreserved veins and arteries. The complication percentages also were shown to be dependent on the region of implantation, vessel type, and vessel source region. Other related risks were brought to attention with allograft options such as steal syndrome and venous hypertension. (Harlander-Locke et al., 2017). While these data points are liable to the great degree of variability in the condition and pathology-progression among individual subjects, as well as their varying region-of-implantation of the vascular grafts, the relative challenges associated with utilising allograft vessels is effectively demonstrated, while simultaneously indicating the potential for treatment processes such as cryogenic freezing to increase viability, hinting potential in these methods.

Xenografts - Xenograft vasculature is not employed as much as other suitable graft options for clinical usage. Nonetheless, revised xenograft conduit options are continually being developed and evolved such as the SynerGraft® Vascular Graft Model 100 (SGVG 100, CryoLife, Inc., Kennesaw, GA, USA) (Darby et al., 2006; Pashneh-Tala et al., 2016) which had shown potential viability in preclinical models for haemodialysis vascular access. These depopulated bovine uretic xenografts are differentiated by the fact that there are no chemical treatment methods used in conditioning, to allow neo cellularisation by the host and encourage retention of native biomechanical properties. In the 23 patients, as per follow up assessments 370±256 days post-surgery, SGVG 100 grafts had returned 12 instances of occlusion, 14 instances of stenosis and 2 infections, compared to 3 PTFE conduits which became infected by Staphylococcus intermedius (epidermis) – however, infections can likely associate with the surgical process as opposed to the efficacy and susceptibility of the conduits themselves. Irrespective of occlusion events however, 12-month patency was comparable to PTFE conduits within 1.2% showcasing some similarities between the performance of the conduits in particular domains. Xenograft methods however don't appear to be pursued as strongly as others. Substantial structural similarity between human and animal vessels is a strong proponent for investigation into these strategies, but revascularizing small vascular tissues such as the layered structure of a blood vessel with the human-derived cell counterparts has proven a massive practical challenge (Ma et al., 2017), and in their current forms offer very little advantage over artificial alternatives (Pashneh-Tala et al., 2016; Si, 2018).

Artificial Conduits

In the absence of viable autologous grafting options, continual and ongoing developments to the manufacturing, treatment, conditioning, and cellular processes involved with developing ABV conduits guides research towards artificial solutions. ABV fabrication methods can include Paste and Expansion, Electrospinning, Thermal-Induced Phase Separation, Braiding, 3D printing, Hydrogel polymerisation, Gas Foaming, (Wang et al., 2020), rapid prototyping, lost wax, dip coating, selective dissolution, and salt leaching (Uchida et al., 2008). Attempts to make use of biological mechanisms e.g., angiogenic methods, hold quite variable success (Guo et al., 2017).

PU and PCL-based conduits - PCL conduits are one of the most used synthetic vascular graft polymers, it degrades in such a way that by around 18 months post-implantation, as emulated in animal models, loses 70-80% of its original weight. The major proponents for these crucial remodelling events are attributed to calcification and stenosis of the conduit, however, these changes have been observed as early as 1.5 months post implantation (Moore et al., 2022). As is common practice in the research space, the combination of various materials offers avenues for

changing the properties of artificial solutions. In recent sheep carotid models for example, combined PCL/chitosan grafts showed improved remodelling, having produced a fully endothelialized lumen surface, organized smooth muscle cells, and the deposition of elastin and collagen like that of native sheep arteries. After the 6-month evaluation period, only 9.1 ± 5.4% of the original PCL/chitosan grafts remained showing that the addition of chitosan had accelerated degradation (Hu et al., 2022; Moore et al., 2022). PU has been routinely recognised as a potential material for artificial vessels because of its attractive mechanical and biocompatibility properties, which justifies its use in catheters and artificial heart-assisting devices (Nguyen et al., 2013). It is less hydrophobic than PCL, which has proven influential in promoting endothelial proliferation, however PU vessels are subsequently more prone to thrombosis (Uttayarat et al., 2010). The biodegradability and hydrophobicity of PCL justifies its combination with PU in many ABVs. Other strategies which have been combined with PU materials entail the use of angiogenic factors, e.g., scaffolds loaded with SDF-1α, VEGF, and cytokines, but encounter challenges with reducing occlusion and longer term (>6 month) patency.

PTFE [/less degradable] -based conduits - The fact that PCL conduits characteristically degrade in time, distinguishes them largely from applications of the other commonly used polymers for vascular conduits, like ePTFE. The stability and inertness associated with ePTFE and similar inert polymers such as PET has been attributed by the literature at least in-part to the electronegativity of their surface, functioning to inhibit deterioration in vivo and minimising reactions with blood components (Wang et al., 2020). In larger diameter vessel applications, ePTFE and PET-based conduits perform well, even in long term usage. However, since mechanical needs, biological interactions and fluid dynamics interact differently with these conduits at smaller diameters, such success at larger scales does not translate to the smaller. These alternatives tend to fail due to occlusion and thrombosis at lower blood flow rates, and chronic foreign body response around the graft are thought to limit vessel functionality and promote hyperplasia (Geelhoed et al., 2017; Wang et al., 2020). For years these have served as the base components of many ABVs, although research tends to show growing interest in natural components with hopes of offering necessary structural and morphological characteristics through manufacturing and treatment processes.
For instance, in a twenty-four weeklong study, the relative successes were illustrated, of an acellular, silk-derived ABV across forty-three 8-week-old male rats, twenty-two of which were implanted with the (*Bombyx mori*) electro spun silk conduit while the remaining twenty-one were implanted with an ePTFE based conduit (Filipe et al., 2018). These conduits were sutured in place of a portion of the abdominal aorta. At the twenty-fourth week, ePTFE-implanted rat survival rate was 73% compared to the 95% survival rate of silk-implanted rats. Not only was thrombogenicity integrally lower with the silk scaffold, but turned out to be 7.6 times more elastic than purely ePTFE counterparts, more closely resemble native rat aorta vessel tensile strength, and although exhibited a lower burst pressure than native vessels, outperformed most other synthetic graft material burst pressures (Wu et al., 2012). Cellularised conduits (Tiwari et al., 2002) also describe ongoing compromises in the application of a tissue-engineered cardiovascular grafts, needing weeks to culture scaffolds, and often requiring the excessing of a substantial portion of seeded cells as they are striped from the scaffold with the introduction of physiological blood flow. This is a common deterrent for the development of cell-seeded conduits, since not only has integration proven a delicate challenge, but by extension, managing adequate cellular proliferation of the vessels longer-term while simultaneously preventing hyperplasia is a balance that cell-seeded conduits especially must achieve for viability.

PROJECT INNOVATION

Solution Description

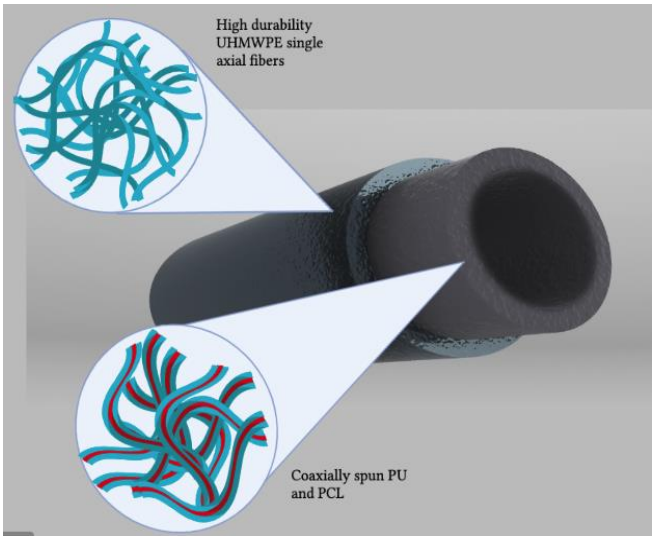


Figure 1: Simple schematic of two-layer conduit, showing coaxially spun inner wall and homogenous uniaxial outer

A two-layer electro spun conduit (Figure 1). The inner layer is made of coaxially electro spun PU and PCL. The superficial layer of PU facilitates effective attachment as per characteristic PU properties, and as it degrades in time, the sub-seated layer of PCL beneath is exposed, the hydrophobicity of which functions to controllably slow cell proliferation, inhibiting hyperplasia and thrombosis. The outer layer is made of a more loosely-electro spun Ultra-High Molecular Weight Polypropylene, a strong, functionally hydrophobic, biocompatible polymer which prolongs the viability/ durability of the scaffold, aided in this by its lubricative properties. "Loosely electro spun" refers to the fact that this layer is (by changing collecting drum rotational speed) spun less tightly than the inner, which facilitates a degree of circumferential dilation in the conduit during changes in blood pressure and volume, to decelerate wear. UHMNWPE is widely employed in medicine, FDA-approved as an implantable, and is characterised by its very high-wear resistance, ductility, and biocompatibility (Hussain et al., 2020) and for these reasons, serves long term function.

Consideration/ Indications per Literature	**Affirmed and/or achieved in…**	**Influence in scaffold solution**
Pore sizes between 5-150 um diameters to allow nutrient diffusion and facilitate adequate cellular adhesion and proliferation. If less than 5um, native tissue granulation is observed, if more than 150um, thrombus formation encouraged.	(Nguyen et al., 2013)	Proponent for electrospinning (as well as out of established practice)
The manufacturing process must entail parameters to control conduit properties depending on implant location	(Wang et al., 2020) Other manufacturing methods: (Uchida et al., 2008)	Strong proponent for electrospinning
Withstand physiological burst pressure ranging (location dependent) from 869mmHg to 4195mmHg, mechanical strength ranges between 1-3 MPa	(Uchida et al., 2008) (Neufurth et al., 2015)	Inclines use of UHMWPE
Balancing endothelialization and inhibiting anti-platelet adhesion in polymer scaffolds – hence, balance between hyperplasia/occlusion and cell adhesion and proliferation	(Nguyen et al., 2013; Tiwari et al., 2002)	Justifies the coaxial electro spun inner wall
Hydrophilicity tends scaffolds towards effective adhesion, however eventual thrombosis, Hydrophobic materials can counteract these tendencies.	(Uttayarat et al., 2010)	Use of PU and PCL in coaxial deign – allow delicate control of cell proliferation
In cellular conduits – cell seeding, culturing, proliferation – are subject to prolonged, potentially costly, often inefficient, and complicated preparation – with highly variable long-term patency	(Guo et al., 2017; Ma et al., 2017; Tiwari et al., 2002)	Deterrent for cellular seeding – however inspired potential lubricant loading

Integrating natural materials has generally tended conduits towards better proliferation, yet increases the rate of degradation	(Hu et al., 2022; Moore et al., 2022)	Conflict of interests between proliferation and degeneration – long term viability was more preferred

Figure 2: Table of solution considerations and influence on design process

Promise and Innovation

The primary points of innovation regard three properties. Firstly, the purposefully dual-layer (inner and outer) electro spun design distinguishes it greatly from other artificial conduits which are traditionally of a single-layered homogenous composition. Secondly, the use of coaxial electrospinning as a strategy to respond to the [time-dependent] changing requirements of vascular conduits in vivo (e.g., from initially having to facilitate strong attachment, and later needing to inhibit hyperplasia) has not been widely employed in the research space - and to the immediate awareness of this report, not at all. Thirdly, in response to the most outstanding shortfall of small-diameter ABVs, UHMWPE, a polymer characterized by its exceptional biocompatibility, ductility, strength, and wear-resistance is being utilized as the long-term outer layer of the conduit. Considering such attractive material properties, it is not exactly clear why past ABV designs have not considered its use for such a vital application. Since most elements of the design model are established and have been re-evaluated over decades, the model bears great promise in offering a longer-term, biocompatible, replicable, and tailorable small diameter ABV conduit.

PROJECT DESIGN (1)

MANUFACTURING [Few days] - The manufacturing process entails the polymerisation of PCL, PU and UHMWPE respectively. Since PU and PCL is electro spun in a coaxial manner, a coaxial compatible syringe/ needle must be used. The PCL solution is loaded into the core channel of the siring, while PU is loaded into the shell channel. Parameters: voltage of 17kV, flow rate of of 25 g/ 10 mins, a collector distance of 20cm and a rotary speed of 2600rpm. This then electro spun until it reaches a tubule wall thickness of 1 ± 0.1mm. After this, the single-axial syringe which has been loaded with polymerised UHMWPE is substituted in place of the coaxial syringe, the angular speed of the collecting drum is lowered to 2400rpm, and electrospinning recommences until conduit wall thickness is 2 ± 0.1mm. Note that the conduit diameter and wall thickness is variable depending on the implantation location, controlled by the collector diameter – placeholder value provided.

IMPLANTATION [Day Surgery] - This ABV is comprised of materials and manufacturing methods which are by no means foreign to the field of application, so it is not expected that specialised training or any alteration of the implantation process should be necessary. The conduit is designed to function alongside the well-established methods of suturing blood vessels.

ASSESSMENT AND PATENCY [12 months post-implantation] - Assessment will take place before, during, and after implantation in preclinical/ clinical models. Testing before the conduit is deemed fit-for-use is necessary and is conducted in practice and in assessment.

BEFORE	DURING	AFTER
Quality control measures: • Appropriate lumen and conduit diameter • Wall integrity	Cross-sectional histological analysis observing adhesion, cell distribution, degree and nature of occlusion, and changes in conduit morphology	Cross-sectional histological analysis observing adhesion, swelling, cell distribution,

• Assure no leakage • [5-150um] Fibre thickness and pore size measure (SEM)	• 2 weeks • 4 weeks • 3 months • 7 months	degree and nature of occlusion, and changes in conduit morphology at approximately 12 months

Figure 3: Table describing the stages and nature of conduit assessment throughout project.

Considering the potential for commercial variation of the conduit utilising other materials, e.g., substituting out UHMWPE, polymers with varying properties can be loaded with lubricating factors, such as Stearic Acid. Lubricating ePTFE with chemical, mechanical and even thermal alternatives are recognised as valid mechanisms to improve durability (Ranjbarzadeh-Dibazar et al., 2014). The outer layer of UHMNWPE would be the last layer of conduit to undergo major degradation, expected to maintain adequate form within in the timescale of about 12 months, however admittedly UHMWPE can have quite variable rate, nature/ cause, and undergo different types of degradation over time depending on implantation conditions (Hussain et al., 2020).

PROJECT OUTCOMES AND BENEFITS – Shortfalls and Future Directions

To be clear, the ABV model proposed is by no means an ideal conduit solution. What might characterize a perfect ABV, specifically small diameter vessels, is one which invokes attachment of native cells (or have been seeded with cells), developing a three structured wall: an intima, media, and adventitia – mimicking native vessels. It should function for the stimulation and support of native vessel repair. The pitched conduit simply advances existing conduits towards these ends. A substantial level of development and design reassessment is foreseeable and expected to be a continual and prolonged process. Implantable designs have potential for immunogenicity, inflammation, infection, patient-specific reactions and, if one may deem it such, another shortfall may be that there are no cellular or chemical factors used to reduce these potential responses imposed by the body, which is another noteworthy avenue of future development.

Value to the Market, Research, and Medicine Spaces

In contrast to the existing treatment solutions, some valuable response attempts to the most prominent obstacles in the medical area are provided. For instance, the multilayered conduit degrades functionally in time, employing bio-compatible polymers which render the possibility of longer term (>6 months) ABV patency. Since the coaxial layer of the vessel utilizes two polymers which have relatively counteracting tendencies regarding occlusion, the concentration of these polymers can be altered in manufacturing to find a balance between hyperplasia and inertness – an un-attempted strategy for maintaining vessel viability and encouraging reendothelialization. This conduit advances ABV viability through manufacturing methods, long term patency through materials and microarchitecture, and retains cost efficiency and high replicability through use of established methods. If not achieved in this exact design, what such a design model may give rise to is the necessary emphasis on long term patency for ABVs – which entails the avoidance of immune-invoking conduits, as well as exploring new methods by which long term patency may be achieved. Other value is held not only in the solution itself, but also in the function of health preservation. Artificial implants often function as temporary management for patients on medical waiting lists, e.g., offering atherosclerotic patients avenues for quick (as is the timescale for such conditions) and cheap vessel conduits, without risking donor site infections, and likelihood of life-threatening occlusive complications. In all, while existing solutions are undoubtably far from flawless, deliberate steps towards an ideal solution require innovative and new ideas to come to fruition in the form of testable, replicable and reliable models like this.

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